INCH-POUND

MIL-P-48043A (AR)
AMENDMENT 4
20 APRIL 1990
SUPERSEDING
AMENDMENT 3
7 June 1988

MILITARY SPECIFICATION

PROJECTILE, 155MM, ILLUMINATING
M485A2, METAL PARTS ASSEMBLY

This Amendment forms a part of Military Specification MIL-P-48043A (AR), dated 26 October 1983, and is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 2.1.2, Other Government documents, drawings, and publications: Add the following new inspection equipment drawing:

"8593429 - Band Seating Testing Machine Indentor and Test Data"

PAGE 3

3.2 Heat treatment: Change second sentence:

From: "No work involving plastic flow is permitted after tempering."

To: "No work involving plastic flow is permitted after tempering unless followed by a stress relief".

PAGE 4

* 3.5.4, Seating. Change the paragraph to read as follows:

From:

"3.5.4 Seating. The diametrical clearance between the rotating band and the band shall not exceed .006 inches when the destructive method of testing is used, or .009 inches when the non-destructive method of testing is used."

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To:

"3.5.4 Seating. The diametrical clearance between the rotating band and the band shall not exceed .006 inches when the destructive method of testing is used, or .016 inches when the non-destructive method of testing is used."

PAGE 7

4.3.1, Submission: Change the last paragraph:

From:

"Five (5) obturating band molded samples made in accordance with ASTM-D638, Type 1 specimens with a thickness of 0.125 ± 0.015 to test the plastic material used to injection-mold or extrude the obturator bands. The specimens and obturator bands shall be moisture-conditioned and heat-sealed in plastic polyethylene bags to retain the moisture."

To:

"Five (5) obturating band molded samples made in accordance with ASTM-D638, Type 1 specimens with a thickness of 0.125 ± 0.015 inches to test the plastic material used to injection-mold or centrifugally cast the obturator bands. The specimens and obturator bands shall be moisture-conditioned and heat-sealed in plastic polyethylene bags to retain the moisture."

PAGE 8

TABLE I, First article inspection: Add the following new inspection to the Base Plug (Dwg 9214146):

"Macro Etch	2	3.1	4.5.1.3"

TABLE I, First article inspection: Add the following new inspections to the Body Assembly (Dwg 9362659):

"Rotating Band bandseat clearance	2	3.5.4	4.5.2
Sidewall contact	2	3.5.3	4.5.2.3
Mechanical Properties of rotating band	2	3.5.5	4.5.2.4"

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PAGE 11

4.4.2.1, Obturator Band: Delete Major 102 in its entirety without substitution:

"102 Width	0.40%	3.1	Gage"

PAGE 12

4.4.2.2, Base Plug: Change Major 105:

From:

"105 Underside of flange perpendicular with press fit diameter	0.40%	3.1	Gage"

To :

"105 Runout of underside of flange with press fit diameter	0.40%	3.1	Gage"

4.4.2.2, Base Plug: Add new Major 108:

"108 Macro Etch	*	3.1	4.4.3.1.3/ 4.5.1.3"

PAGE 14

4.4.2.3, Projectile Body, Prior to Banding, change Major defect 125:

From: "125

Runout of obturator band seat with bourrelet diameter	0.40%	3.1	Gage"

To :

"125 Runout of rear face with bourrelet diameter	0.40%	3.1	Gage"

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4.4.2.3, Projectile Body, Prior to Banding, add new Major defect 126:

	"126	Runout of obturator band seat with bourrelet diameter		0.40%	3.1 Gage"

PAGE 15

4.4.2.3, Projectile Body, Prior to Banding, change Minor defect 208:

From:	"208	Surface finish in press fit area		0.65%	3.1 Gage"
To:	"208	Surface finish in press fit area		0.65%	3.1 Visual"

PAGE 16

4.4.2.4, Rotating Band Blank, change Major 102:

	From:	"102	Thickness	0.40%	3.1 Gage"
*	To:	"102	Thickness (minimum)	0.40%	3.1 Gage"

4.4.2.4, Rotating Band Blank: Add new Minor 202:

*	"202	Thickness (maximum)		0.65%	3.1 Gage"

PAGE 17

4.4.2.5, Body Assembly (after painting): Delete "4.4.2.5" in its entirety and substitute new "4.4.2.5" included.

PAGE 18

4.4.2.6, Projectile Metal Parts: Change Major 101:

From:					
	"101	Rotating band damaged		0.40%	3.1 Visual
To :					
	"101	Rotating band damaged	**	0.40%	3.1 Visual"

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF	DRAWING NUMBER
4.4.2.5	Body Assembly (after painting)			9362659
				NEXT HIGHER ASSEMBLY
				9214142

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of bourrelet, max.		100%	3.1	Gage
102	Salt spray		*	3.1	4.5.3
Minor					
201	Protective coating inadequate or damaged		0.65%	3.1	Visual
202	Paint on threads		0.65%	3.11	Visual
203	Paint on base fit area		0.65%	3.11	Visual
204	Paint overspray on rotating band exceeding allowable limit		0.65%	3.1	Gage
205	Paint overspray on press fit diameter exceeding allowable limit		0.65%	3.1	Gage
206	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES: * Sampling size as specified in applicable test method paragraph.

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4.5.2.3, Sidewall contact: Change:

From:

"4.5.2.3 Sidewall contact. Samples provided by 4.4.3.2.3 shall be visually examined for scale formation on the surface which was adjacent to the seat and for contact with the sidewalls of the band seat."

To :

"4.5.2.3 Sidewall contact. The samples provided in accordance with 4.4.3.2.3 shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward and rear of rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other section shall be sectioned into two parts, thus exposing four surfaces each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements."

4.5.2.4, Mechanical properties: Change:

From:

"4.5.2.4 Mechanical properties. The sample provided by 4.4.3.2.4 shall have the rotating band removal and carefully flattened without hammering so as to hold additional cold working to a minimum. A test specimen conforming to a sheet type, 1/2 inch wide, rectangular tension test specimen as prescribed by ASTM E-8 shall be prepared and tested in accordance with the procedures prescribed by ASTM E-8."

To :

"4.5.2.4 Mechanical properties. The sample provided by 4.4.3.2.4 shall have the rotating band removed and carefully flattened without hammering so as to hold additional cold working to a minimum. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen shall be removed from the blank of a one inch gage length and one-half inch width and grip ends of sufficient length to accommodate the tensile machine and then tested in accordance with the procedures prescribed by ASTM E-8."

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4.5.7.1, Excess pressure: Change:

From:

"4.5.7.1 Excess pressure. Each projectile submitted for this test shall be assembled with an inert loaded candle, a dummy fuze, and an inert expelling charge. Each projectile will be fired from an M199 or M185 cannon tube with an M119A1 propelling charge weight adjusted and temperature pre-conditioned at +145 degrees F to yield a chamber pressure of 115 ± 2% of peak pressure. An attempt will be made to recover the projectiles fired for this test. As an alternative, the M185 gun tube with the M119A2 Prop Charge may be utilized for the performance of the excess pressure test phase.

* To :

"4.5.7.1 Excess pressure. Each projectile submitted for this test shall be assembled with an inert loaded candle, a dummy fuze, and an inert expelling charge. Each projectile will be fired from either an M199 or M185 cannon tube with an M119 series propelling charge. The propelling charge shall be weight adjusted and temperature pre-conditioned at +145 -5 degrees F to yield a chamber pressure of 42,000 p.s.i. to 46,000 p.s.i.. Test pressures outside the prescribed range will be referred for disposition by the technical agency. An attempt will be made to recover the projectiles fired for this test.

4.5.7.2, Service pressure: Change first sentence:

From:

"Each projectile submitted for this test shall be assembled with an inert loaded candle, an M565 Fuze and an expelling charge."

To :

"Each projectile submitted for this test shall be assembled with an inert loaded candle, an M565 or M577 Fuze and an expelling charge."

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4.4.2.7, Pallet: Add the new "SPECIAL" section between the "CRITICAL" section and the "MAJOR" section:

"SPECIAL

a. -Cover support missing (8 places) 100% 3.1 Visual"

4.4.2.8 Add new subparagraph "4.4.2.8", (included)

* Included as page 19a of MIL-P-48043

PAGE 20

4.4.3.1, Mechanical properties: Add new paragraph "4.4.3.1.3" after paragraph 4.4.3.1.2:

"4.4.3.1.3 Macro Etch (base plug). Sample test quantities of base plugs shall be subject to a Macro Etch Test to comply with the cross forging requirement in accordance with 4.5.1.3."

PAGE 21

4.4.3.3.1 Add the following sentence to the beginning of the paragraph:

"All obturating bands shall be moisturized by boiling in water until 2.0% to 3.0% moisture by weight is attained prior to assembling the projectile".

PAGE 22

4.5.1, Mechanical Properties: Add new subparagraph 4.5.1.3

"4.5.1.3 Macro Etch (base plug). One base plug from each standard heat treatment lot and one from each miscellaneous heat treatment lot shall be subjected to a Macro Etch Test. The base plug shall be sectioned through its diameter and one of the sectioned surfaces prepared by grinding the surface to a flat. This part shall then be immersed in a macro etch solution of 50 percent hydrochloric acid and 50 percent water and heated to a temperature of 180 degrees F, minimum. The base part shall remain in the macro etch solution until flow lines become clearly visible. The part shall then be rinsed with water, dried and visually examined for compliance with dwg 9214146."

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4.5.7, Proving ground ballistic test: Change:

From:

"The following tests shall be conducted using cannon tubes having 25 percent remaining life based upon current gun tube condemnation criteria. Photographic equipment shall be positioned near the muzzle of the weapon to record the condition of the projectile as it exists the tube. Chamber pressures and velocities shall be recorded for each round fired. All photographs and recovered hardware shall be examined for any noncompliance with the applicable requirements of 3.9. Range, deflection and velocity shall be recorded for each round fired for information."

To :

"The following tests shall be conducted using cannon tubes having 25 percent remaining life based upon current gun tube condemnation criteria. Photographic equipment shall be positioned near the muzzle of the weapon to record the condition of the projectile as it exists the tube. Chamber pressures and velocities shall be recorded for each round fired. All photographs and recovered hardware shall be examined for any noncompliance with the applicable requirements of 3.9. Range, deflection and velocity shall be recorded for each round fired for information. An attempt will be made to recover all projectiles fired per paragraph 4.5.7.1 (excess pressure). For those projectiles fired per paragraph 4.5.7.2 (service pressure). For those projectiles fired per paragraph 4.5.7.2 (service pressure), a recovery shall be attempted only if an unusual firing occurs in any zone. Examples of unusual firings include, but are not limited to, erratic flight, projectile outside round impact area, fuze malfunction, or indication of metal parts separation."

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4.4.2.6, Projectile Metal Parts: Change Major 102:

From:
"102 Obturator missing or damaged 0.40% 3.1 Visual"

To :
"102 Obturator missing or damaged **0.40% 3.1 Visual"

4.4.2.6, Projectile Metal Parts: Change Major 104:

From:
"104 Threads damaged 0.40% 3.1 Visual"

To :
"104 Threads damaged ** 0.40% 3.1 Visual"

4.4.2.6, Projectile Metal Parts: Change Major 105:

From:
"105 Weight 0.40% 3.1 Scale"

To :
"105 Weight **0.40% 3.1 Scale"

4.4.2.6, Projectile Metal Parts, change Minor 202 to read as follows:

"202 Burr ** 0.65% 3.11.1 Visual"

4.4.2.6, Projectile Metal Parts: Change Minor 203 to read as follows:

"203 Foreign matter ** 0.65% 3.11.2 Visual"

4.4.2.6, Projectile Metal Parts: Add new note to the notes:

"** Inspections to be performed prior to assembly of the lifting plug and grommet."

PAGE 19

4.4.2.7, Pallet: Change drawing number:

From: "7549275"

To: "8837839"

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Body Assembly (after banding)		9362659
			NEXT HIGHER ASSEMBLY 9514142

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Magnetic particle stamp missing (kndicating failure to pass test)		100%	3.1	Visual
Major					
101	Runout of rotating band with bourrelet and internal base diameter		0.40%	3.1	Gage
102	Diameter of rotating band		0.40%	3.1	Gage
103	Diameter of rotating band groove		0.40%	3.1	Gage
104	Length from rear face to rear shoulder of rotating band		0.40%	3.1	Gage
105	Rotating band bandseat clearance		*	3.5.4	4.4.3.2/4.5.2
106	Sidewall contact		*	3.5.3	4.4.3.2.3/4.5.2.3
107	Mechanical properties of rotating band		*	3.5.5	4.4.4.2.4/4.5.2.4
Minor					
201	Width of rotating band		0.65%	3.1	Gage
202	Depth of undercut front and rear of rotating band (if applicable)		0.65%	3.1	Gage
203	Width of undercut front and rear of rotating band (if applicable)		0.65%	3.1	Gage
204	Radii or chamfer missing or incorrect		0.65%	3.1	Visual
205	Width of groove in rotating band		0.65%	3.1	Visual
206	Evidence of poor workmanship		0.65%	3.11	Visual

NOTE: * Sampling size as specified in applicable test method paragraph.

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5.2.1.1, Projectile body: Change:

From: "7549275"

To: "8837839"

PAGE 28

5.3, Marking: Change:

From: "7549275"

To: "8837839"

PAGE 29

6.7, Moisturizing of obturating bands: Delete in its entirety without substitution.

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